

Zaid Ahmad

Ottawa, ON | (613) 262-2906 | zaidahmad8060@gmail.com | zaidahmad.dev | linkedin.com/in/zaid-ahmad-ba9b10224 | github.com/zaidahmad16

EDUCATION

Carleton University

Ottawa, ON

B.Sc. Computer Science, AI and Machine Learning Stream, Minor in Mathematics

Sept 2024 - May 2029

- **Coursework:** Data Structures and Algorithms, C/C++, Linear Algebra, Discrete Structures II, Web Applications

TECHNICAL SKILLS AND CLEARANCES

Languages: C++, C, Java, Python, JavaScript, TypeScript, Go

Frameworks and Libraries: React (Next.js), React Flow, Node.js, Express, Tailwind CSS, Hugo, BeautifulSoup, scikit-learn

Tools and Platforms: Git, GitHub Actions, Docker, Linux, POSIX Threads, Prometheus, Grafana, Vitest, pytest, Azure DevOps

Databases and Cloud: PostgreSQL, MongoDB, Firebase, Supabase, FastAPI, REST APIs, AWS, Vercel, CI/CD

Concepts: Concurrent Programming, Systems Programming, Networking, Machine Learning, Database Schema Design, Unit Testing

Clearance: Government of Canada Reliability Status Security Clearance, Issued 2026

EXPERIENCE

Business Analyst (FSWEP)

June 2026 - Aug 2026

Immigration, Refugees and Citizenship Canada

Ottawa, ON

- Intake spanned case-management and non-case-management requests across four categories. Triaged submissions in Azure DevOps and classified scope and category (Cloud, AI, Software, Potential Project), standardizing dashboard entries covering business need, impacts, and consultation.
- Business owners submitted generic, unquantified impact claims. Issued targeted clarification correspondence and flagged unsupported claims as follow-up gaps, converting input into quantified statements on transaction volumes, affected client records, and multi-system scope.
- Approved business needs had to reach review as auditable requirements. Authored High-Level Business Requirements, including for a governed low-code application platform, self-auditing each draft into numbered capability statements traceable to source.
- Recurring annual change requests returned every governance cycle. Updated precedent figures, dates, and references instead of redrafting, routed qualifying items to an expedited path, and reconciled the departmental prioritization list after each review, cutting drafting turnaround.
- Drafting conventions lived in individual practice, not in writing. Codified business need structure, impact standards, official-languages requirements, and a pre-send checklist into a shared reference, improving consistency of team intake output.

Software Developer (Volunteer)

May 2026 - Aug 2026

Carleton Computer Science Society

Remote

- The course-graph Explorer had no way to find a course without scanning the graph. Implemented a type-ahead search in React and TypeScript with normalized code and title matching, reusing the existing store so results highlight identically to a node click.
- The codebase ran tests without reporting coverage. Integrated @vitest/coverage-v8 and configured HTML and terminal reports in vite.config.ts, landing through every lint, format, and typecheck CI gate.
- Explorer users could not narrow the graph to core courses. Implemented a React and TypeScript toggle deriving the visible node set outside the layout engine and recomputing dagre layout on state change, establishing a reusable filter seam.
- The webview needed an explorable visual entry point. Built a force-directed d3-force and SVG view with drag, pan/zoom, fit-to-view, theme tokens, and prefers-reduced-motion, exposing a dormant highlight API for future search.

Logistics Lead

Sept 2025 - Present

cuHacking

Ottawa, ON

- A 500+ attendee hackathon needed venue, equipment, and day-of coordination. Led a 2-person logistics team across setup and live operations.

PROJECTS

CarletonCourseMap

carletoncoursemap.ca/map | github.com/zaidahmad16/CarletonCourseMap

Python, Parsel/XPath, FastAPI, PostgreSQL, Next.js, React Flow, dagre, Tailwind CSS, Railway, Cloudflare

- Carleton students planned degrees by reading prerequisite text course by course. Scraped Carleton Central with Parsel/XPath, modelled AND/OR prerequisite logic as a DAG, and stored course data as JSONB in PostgreSQL on Neon, eliminating manual planning for 1,700+ students.
- Scraped course, program, department, prerequisite, and instructor data had to stay consistent. Designed a normalized 8-table PostgreSQL schema with foreign key constraints and JSONB columns, GIN-indexed for fast prerequisite lookups.
- Prerequisite conflicts across a full degree plan were invisible in a course list. Rendered four-year dependency graphs in React Flow and dagre with color-coded nodes and monthly-refreshed RateMyProfessors ratings, surfacing conflicts in real time.
- Serving the graph publicly exposed the data to open traffic. Built a FastAPI backend with rate-limited, CORS-protected endpoints, batch lookup of 100 records per round-trip, and API-key-gated admin routes for curating graph layouts.

RPi Network Packet Analyzer

github.com/zaidahmad16/rpi-packet-analyzer

C, Python, scikit-learn, Raw Sockets, Prometheus, Grafana, AWS, Raspberry Pi 5

- Frame-level analysis ruled out higher-level capture libraries. Built a real-time engine in C on raw AF_PACKET sockets, manually parsing Ethernet, IP, TCP, and UDP headers and tracking per-source traffic with a hash map and min-heap top-K.
- Fixed thresholds flagged ordinary traffic as suspect. Trained a scikit-learn Isolation Forest on captured traffic features and exposed it via a Prometheus endpoint on a Grafana dashboard on AWS EC2, reducing false network alerts.
- The full pipeline had to run on a 4GB Raspberry Pi 5. Deployed capture, detection, and visualization end to end on device, sustaining live analysis under 100MB of memory.